Down Payments, Deferred Homes: How LTV Restrictions Reshape Household Consumption

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The views expressed here are those of the authors, and not necessarily those of the Norges Bank.

Housing and consumption

Housing markets and consumption are tightly linked

- House prices affect consumption via wealth and collateral channels
  (Campbell & Cocco 2007; Mian, Rao & Sufi 2013; Guren et al. 2021)
- This link is one of the main drivers of business-cycle fluctuations and credit-driven booms
  (Bernanke & Gertler 1986; Kiyotaki & Moore 1997; Iacoviello 2005)

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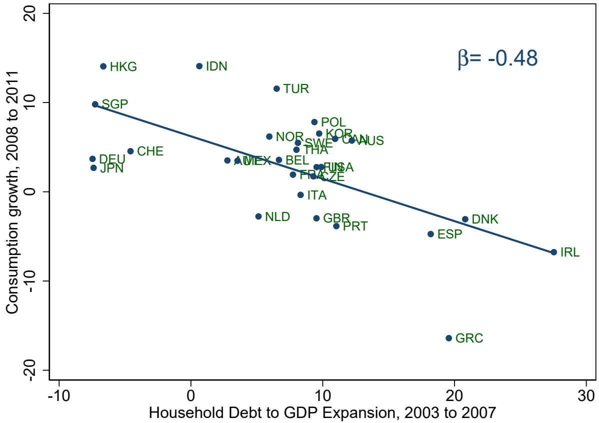
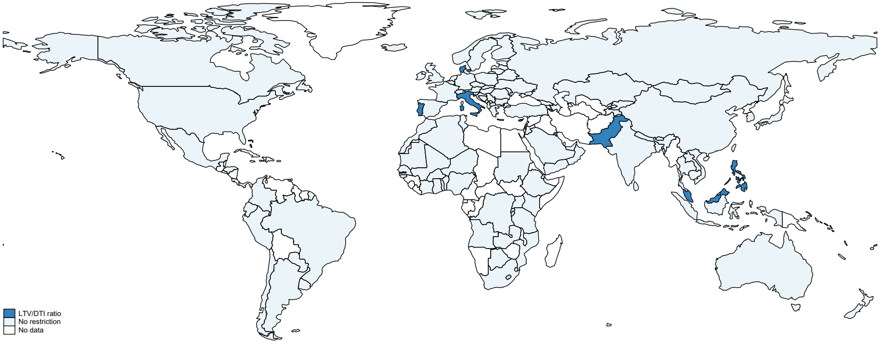
This link is stronger when household leverage is high, amplifying the recessions

Housing and consumption



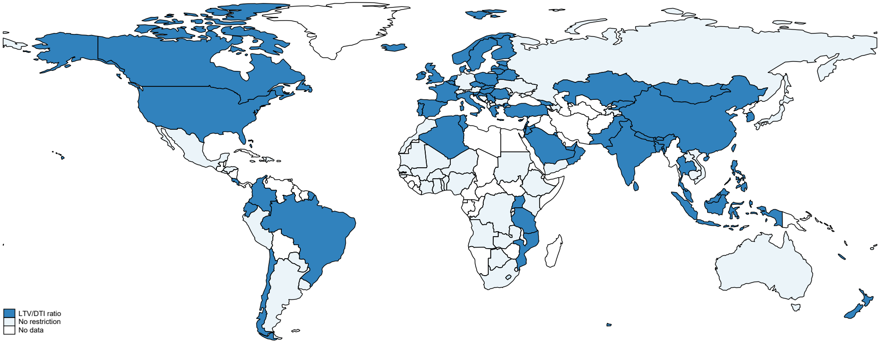
Increase in HH debt before the GFC predicts a decline in consumption during GFC

Macroprudential Policies in 2000



Many countries have implemented borrowing restrictions on HH

Macroprudential Policies in 2021



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Theoretical mechanism

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Important to understand the aggregate demand implications of these constraints

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Standard life-cycle models with binding borrowing constraints

- Tighter LTV $\Rightarrow$ higher down payment
- Renters *increase* savings to meet requirement
- Consumption *falls*

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Models with consumption smoothing and endogenous tenure choice

- Tighter LTV $\Rightarrow$ larger required savings
- Renters *delay* their purchase and saving because they expect higher income in the future
- Consumption *rises*

Balke et al. 2024

$\Rightarrow$ **Fundamentally different implications for aggregate demand**

Preview of Findings

- **Renters increase their consumption after the introduction of LTV restriction**
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Preview of Findings

- **Renters increase their consumption after the introduction of LTV restriction**
  $\rightarrow$ 2.7 pp increase in the consumption/income ratio
- **Mechanism**: Delayed home purchase with delayed savings
  $\rightarrow$ Larger effects when earnings potential is high controlling for current income
- **Homebuyers** decrease their consumption before and after a home purchase
  $\rightarrow$ Evidence for rebuilding liquidity buffers

Data & Empirical Strategy

Data

- Consumption, housing transactions, tax filings, individual characteristics
  $\rightarrow$ Consumption: Debit card transactions, digital invoice, direct remittances. Approximately 80% of the card payments. 22 COICOP categories
- Sample: 2006-2018, Annual, Household-level

Data

- Consumption, housing transactions, tax filings, individual characteristics
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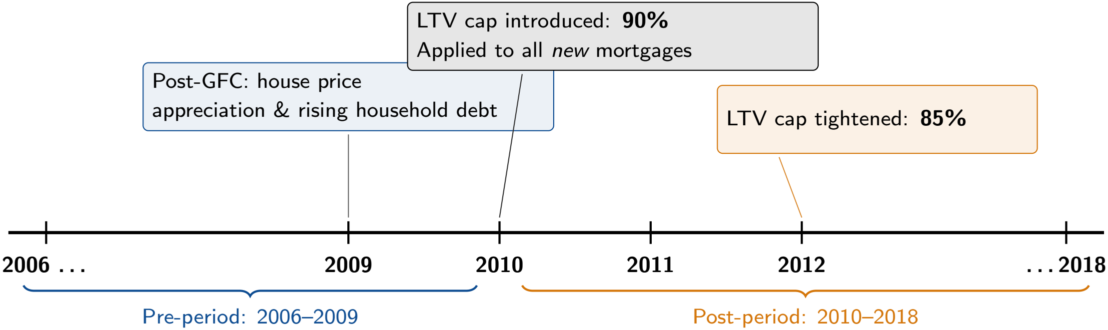
- Sample: 2006-2018, Annual, Household-level
- **3 groups**: Renters, homebuyers, homeowners
  → Renters: Do not have housing wealth, no housing transactions before the restriction
  → Homebuyers: First-time homebuyers
  → Homeowners: Have housing wealth before 2006

The Norwegian LTV Restriction



- LTV regulation introduced in **2010**
- Restriction applies only to *new* mortgages ⇒ existing homeowners unaffected

Difference-in-Differences Design

$$y_{it} = \beta \; Renter_i \times Post_t + \gamma_1 Renter_i + \gamma_2 Post_t + \epsilon_{it}$$

- y_{it} = Consumption / Income
- $Renter_i$: =1 if HH is a renter until 2010, 0 if HH is already a homeowner prior to 2006
  $\rightarrow$ Renters include HHs that purchase a home after the restriction
- $Post_t$: =1 if year$\geq$2010

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  $\rightarrow$ Renters include HHs that purchase a home after the restriction
- $Post_t$: =1 if year$\geq$2010
- **Identifying assumption**
  1. Without the restriction, the consumption/income difference between renters and homeowners would be the same
     $\rightarrow$ The levels can be different
  2. Homeowners are not affected by the restriction
     $\rightarrow$ The restriction can reduce homeowners' access to credit, or housing wealth

Results

Consumption Reaction of Renters

	Consumption/Income		
	(1)	(2)	(3)
Renter × Post	0.0310***	0.0289***	0.0271***
	(0.00356)	(0.00381)	(0.00341)
Renter	-0.0820***	-0.0799***	
	(0.00275)	(0.00262)	
Post	-0.0149**		
	(0.00613)		
Fixed Effects:			
Year FE		✓	✓
Household FE			✓
Obs.	7,610,365	7,610,365	7,610,365
R^2	0.003	0.006	0.428
Mean(Dependent Var.)	0.660		

Renters increase their consumption after the LTV restriction

Do homeowners form a good control group for renters?

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	Consumption/Income							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Renter × Post	0.0313***	0.0305***	0.0237***	0.0258***	0.0225***	0.0330***	0.0370***	0.0401***
	(0.00449)	(0.00438)	(0.00410)	(0.00414)	(0.00379)	(0.00426)	(0.00352)	(0.00149)
Fixed Effects:								
Household FE	✓	✓	✓	✓	✓	✓	✓	✓
Age × Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Immigration background × Year FE		✓	✓	✓	✓	✓	✓	✓
Education × Year FE			✓	✓	✓	✓	✓	✓
Deposit bins × Year FE				✓	✓	✓	✓	✓
Financial asset bins × Year FE					✓	✓	✓	✓
Total wealth bins × Year FE						✓	✓	✓
Debt bins × Year FE							✓	✓
Parental wealth bins × Year FE								✓
Obs.	7,610,365	7,577,947	7,577,946	7,576,625	7,561,299	7,451,124	7,018,986	5,804,736
R^2	0.429	0.430	0.431	0.433	0.436	0.452	0.489	0.537

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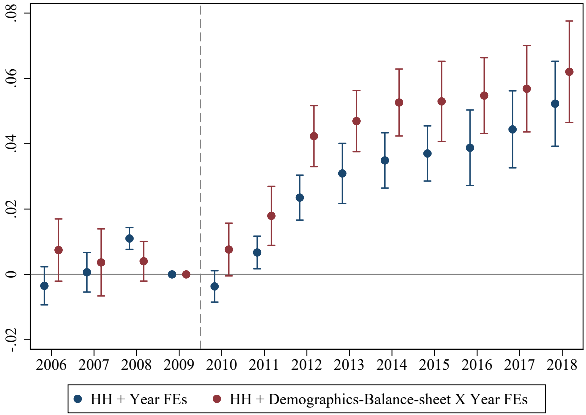

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	Consumption/Income				
	(1) LTV<95%	(2) LTV<85%	(3) LTV<75%	(4) LTV<65%	(5)
Renter $\times$ Post	0.0253***	0.0247***	0.0238***	0.0225***	0.0301***
	(0.00315)	(0.00304)	(0.00290)	(0.00275)	(0.00309)
Fixed Effects:					
Household FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	
Municipality $\times$ Year FE					✓
Obs.	7,239,366	7,071,365	6,818,191	6,453,484	7,610,365
R^2	0.426	0.427	0.429	0.432	0.433
Mean(Dependent Var.)	0.656				

Second home purch.

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3. The effect of homeownership on consumption changes over time
   $\rightarrow$ House-related expenses or ...

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$\rightarrow$ House-related expenses or ...

	Cons. / Inc. − Rent	Cons. / Inc. − Mort. Int.	Cons. − Util. / Inc. − *Util*.	Cons. − Util. / Inc. − Rent − Mort. Int. − Util.
	(1)	(2)	(3)	(4)
Renter × Post	0.0248***	0.0580***	0.0231***	0.0260***
	(0.00274)	(0.00350)	(0.00396)	(0.00397)
Fixed Effects:				
Year FE	✓	✓	✓	✓
Household FE	✓	✓	✓	✓
Obs.	7,610,365	7,610,365	7,610,365	7,610,365
R^2	0.334	0.432	0.355	0.292
Mean(Dependent Var.)	0.767	0.727	0.738	0.939

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3. The effect of homeownership on consumption changes over time
   → House-related consumption or **housing wealth**
   → **Solution**: Renter-only comparison!

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   → **Solution**: Renter-only comparison!

Renter-only comparison

- Are there renters who are less exposed to the restriction?

  → Renters who are very similar to existing homeowners: high ownership capacity
- How do we detect such renters with high ownership capacity?
  1. Split the sample into two: homeowners and renters
  2. Use XGBoost to predict the probability of being a homeowner using 2006 information

  → **Intuition**: **Renters with a high probability** have observables suggesting that they are similar to homeowners but still do not buy. Hence, they **are less affected**.

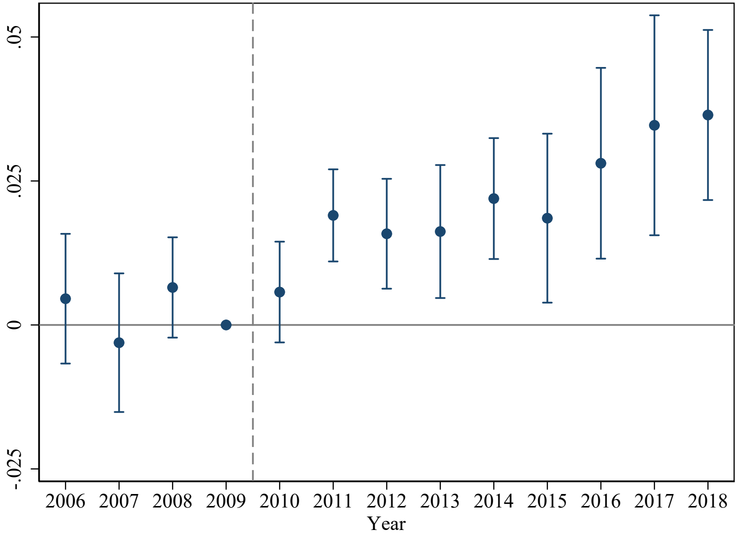
Renter-Only Comparison

$$y_{it} = \beta\ (1 - Homeowner\ \ Prob)_i \times Post_t + \gamma_1 (1 - Homeowner\ \ Prob)_i + \gamma_2 Post_t + \epsilon_{it}$$

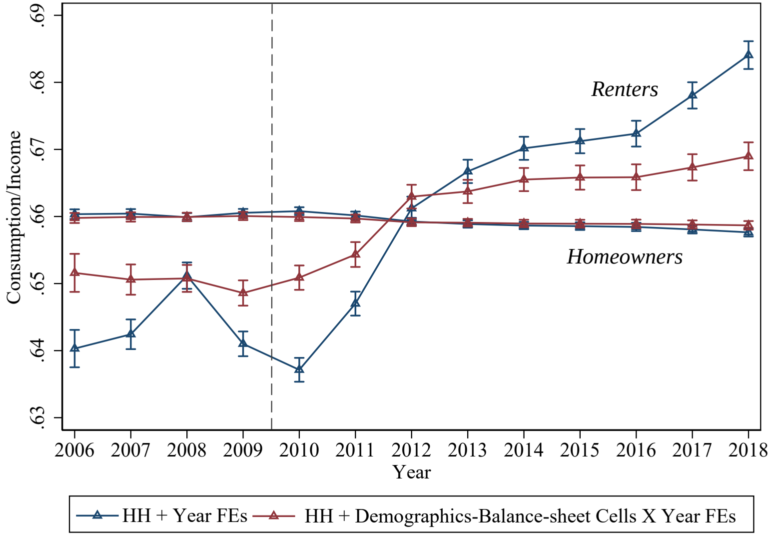
	Consumption/Income			
	(1)	(2)	(3)	(4)
(1 - Homeowner prob.) × Post	0.0589***	0.0466***		
	(0.00986)	(0.0126)		
1(Homeowner prob.<80 $_{pct}$) × Post			0.0340***	0.0197***
			(0.00486)	(0.00507)
Fixed Effects:				
Year FE	✓	✓	✓	✓
Household FE	✓	✓	✓	✓
Matching		✓		✓
Obs.	490,074	128,205	490,074	128,205
R^2	0.464	0.462	0.463	0.462
Mean(Dependent Var.)	0.603			

Results hold when making use of heterogeneity within renters

Renter-Only Dynamic Effects



The Effect is Driven by Renters



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- **Renters may delay their home purchase and savings** (Karlman et al. (2024))
  $\rightarrow$ Renters need to reduce their consumption to accumulate savings
  $\rightarrow$ Leading to a deviation in consumption smoothing, which is costly for renters
  $\rightarrow$ The disutility of this deviation can be larger than the benefits of an earlier home purchase
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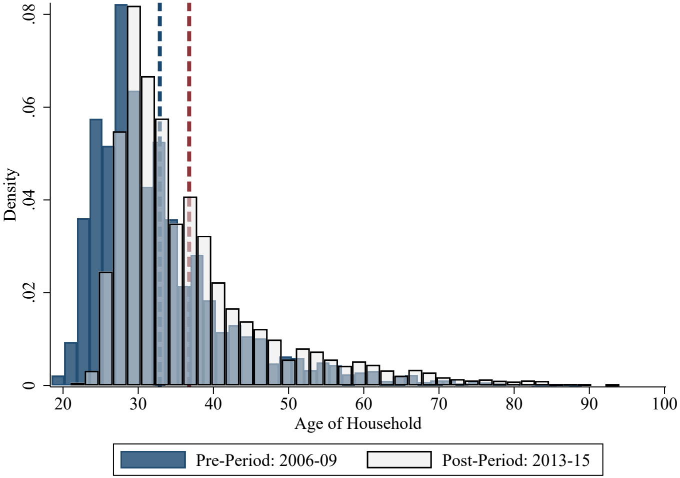
- **Two predictions**
  1. There must be a delay in home purchases
  2. The increase in consumption must be stronger for renters who expect a larger increase in their earnings

Age Distribution of Home Buying Cohorts



Homebuying cohorts get older

Earnings Potential Gap

Idea: How far a household is from the upper end of the income distribution among comparable households who are later in their career stages.

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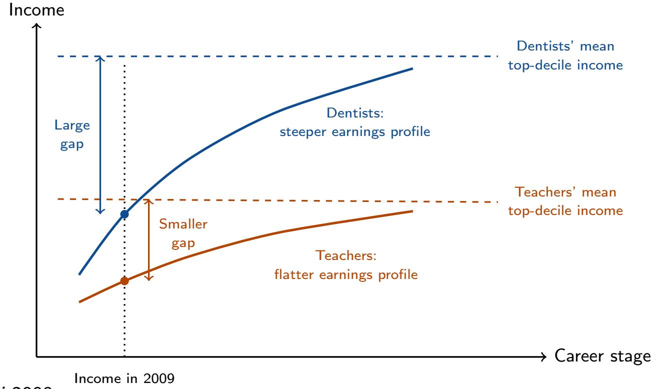


$$\text{Gap}_i = \text{Top-decile income}_c - \text{Income}_{i,2009}$$

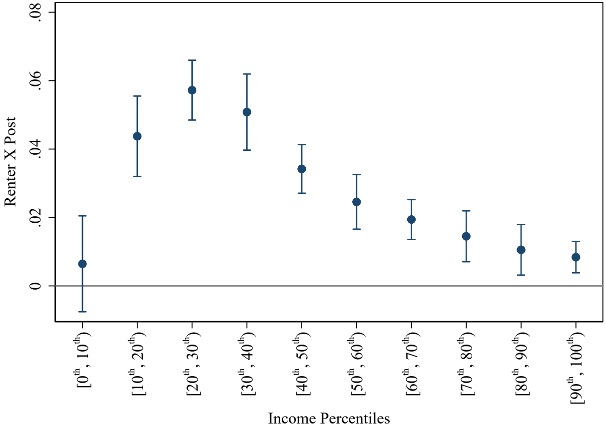
The gap is higher for HHs who are in the early stages of their careers and HHs whose cells have higher frontier income

Earnings Potential Gap

	Above top 10%	1^{st} Quart.	2^{nd} Quart.	3^{rd} Quart.	4^{th} Quart.
	(1)	(2)	(3)	(4)	(5)
Renter × Post	-0.000387	0.00334	0.0124***	0.0255***	0.0376***
	(0.00835)	(0.00374)	(0.00357)	(0.00310)	(0.00316)
Fixed Effects:					
Household FE	✓	✓	✓	✓	✓
Cell × Year FE	✓	✓	✓	✓	✓
Income bins × Year FE	✓	✓	✓	✓	✓
Obs.	622,407	1,549,461	1,608,470	1,614,720	1,620,964
R^2	0.393	0.388	0.420	0.480	0.509
Mean(Dependent Var.)	0.660				

Renters with greater anticipated income growth are more likely to delay the costly deviation from consumption smoothing

Income Heterogeneity



The effect is insignificant for the lowest income decile. But stronger for the rest of low-income renters

Regional Heterogeneity

	MuniLTV Interaction	MuniLTV <25th perc.	MuniLTV 25th–75th perc.	MuniLTV >75th perc.
	(1)	(2)	(3)	(4)
Renter × Post × Municipality LTV	0.00363**			
	(0.00173)			
Renter × Post	0.0300***	0.0204***	0.0280***	0.0353***
	(0.00256)	(0.00448)	(0.00394)	(0.00374)
Post × Municipality LTV	0.0133***			
	(0.00217)			
Fixed Effects:				
Household FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Obs.	7,064,393	1,700,388	3,495,605	1,868,400
R^2	0.429	0.432	0.428	0.427
Mean(Municipality LTV)		0.810	0.921	1.030
Diff. vs MuniLTV < 25th perc.			0.0076	0.0149**
Diff. vs MuniLTV 25th-75th perc.				0.0073

The effect is stronger in regions with higher LTV ratios measured before the restriction

Expenditure Elasticities

	Cons. type/Income		Cons. type/Total cons.	
	(1)	(2)	(3)	(4)
	$\epsilon < 1$	$\epsilon > 1$	$\epsilon < 1$	$\epsilon > 1$
Renter $\times$ Post	0.00346	0.0204***	-0.0193***	0.0214***
	(0.00254)	(0.00120)	(0.00303)	(0.00354)
Fixed Effects:				
Household FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Obs.	7,610,365	7,610,365	7,610,365	7,610,365
R^2	0.579	0.444	0.547	0.542
Mean(Dependent Var.)	0.147	0.380	0.301	0.687

The renters increase their luxury spending, measured by expenditure elasticity

Delayed Savings or Giving-up?

	1(*Labor*)	1(*Business*)	1(*UI*)	1(*Bonus*)	ln(bonus)	1(*Irregular*)	ln(irregular)	1(*Overtime*)	ln(overtime)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Renter × Post	0.0281***	0.00116***	-0.00254***	0.0330***	0.0498**	0.0559***	0.110***	-0.0133	0.0225
	(0.00364)	(0.000277)	(0.000578)	(0.00189)	(0.0222)	(0.00478)	(0.0132)	(0.0110)	(0.0180)
Fixed Effects:									
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Household FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Obs.	7,610,365	7,610,365	7,610,365	7,610,365	642,472	7,610,365	2,198,660	7,610,365	1,605,127
R^2	0.748	0.395	0.268	0.418	0.675	0.549	0.624	0.494	0.588
Mean(Dependent Var.)	0.654	0.008	0.019	0.093	7.304	0.298	7.376	0.219	6.970

No evidence for giving-up, as renters increase their effort in the labor market

Additional Exercises

- The effect decreases as parental income increases
- Locational sorting does not explain the results
- Renters increase their debt and reduce savings (deposits and financial assets)
- Using Δlog(deposits) and ΔDeposits/Income as the outcome variable gives the same results
- Allowing renters to be differentially affected by macroeconomic conditions leaves the results unchanged

What about renters who purchased a home after the restriction?

Homebuyers

Three challenges in estimating the effect of the LTV constraint on homebuyers' consumption

1. Enough time after the restriction but before the purchase to observe pre-purchase behavior
2. Home purchases have a direct effect on consumption (Benmelech et al 2023)
3. The constraint induces a selection into the housing market

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How we address these issues

1. Use renters who purchased a home in 2013 as the treatment group
2. Use a pre-restriction home-buying cohort (2009 cohort) to control for home purchases
3. Match the 2013 cohort to the 2009 cohort to account for selection
4. As these two cohorts are in different years, use homeowners as an additional control group for time effects

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Triple differences design: Compare homebuyers to homebuyers + compare homebuyers before and after the restriction

Homebuyers

	Consumption/Income		(Cons.+Mor.)/Income
	(1)	(2)	(3)
Homebuyer × Prepurchase × Post	-0.0602***		
	(0.00844)		
Homebuyer × Prepurchase	0.000228		
	(0.00398)		
Homebuyer × Post	0.0459***		
	(0.00785)		
Homebuyer × Postpurchase × Post		-0.0177**	-0.0250***
		(0.00734)	(0.00753)
Homebuyer × Postpurchase		-0.0149***	0.0164***
		(0.00468)	(0.00473)
Homebuyer × Post		0.0450***	0.0722***
		(0.00721)	(0.00751)
Fixed Effects & Matching:			
Year FE	✓	✓	✓
Household FE	✓	✓	✓
Matching	✓	✓	✓
Obs.	4,434,946	4,340,866	4,340,866
R^2	0.483	0.500	0.500
Mean(Dependent Var.)	0.658		

Homebuyers lower their consumption before a home purchase and keep it low after the purchase

Homebuyers

	Δ (Dep.+Fin. Assets)/Income$_{t-1}$	Δ Dep./Income$_{t-1}$	Δ Fin. Assets/Income$_{t-1}$	Δ Municipality	1(Home purch.)	log(Muni. income$_{06-09}$)
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.0681**	0.0252**	0.0429***	-0.00122	-0.00186	0.00360
	(0.0264)	(0.0123)	(0.0143)	(0.00296)	(0.00609)	(0.00312)
Sample:						
Matching	✓	✓	✓	✓	✓	✓
Obs.	6,254	6,254	6,254	4,584	6,712	6,712
R^2	0.001	0.001	0.002	0.000	0.000	0.000
Mean(Dependent Var.)	0.062	0.026	0.036	0.012	0.121	12.495

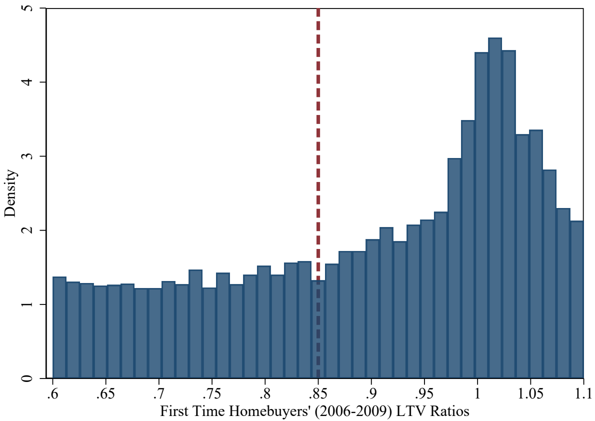
Homebuyers try to rebuild their depleted liquidity buffers

Conclusion

- **Financial stability implications of macroprudential restrictions are not specific to homeowners**
- **LTV restrictions may stimulate short-term demand by increasing renters' consumption**

Thank You!

LTV distribution of first-time homebuyers before the restriction



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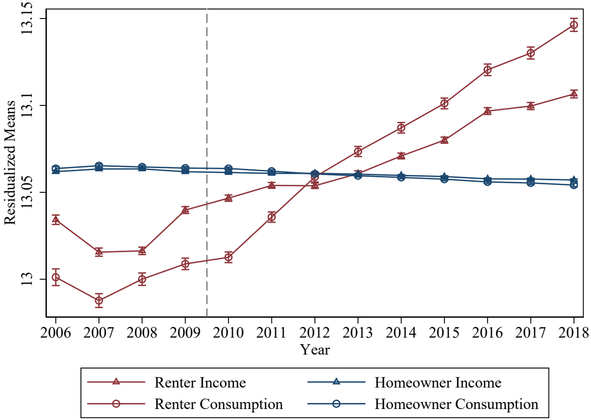
Dynamic Changes in Consumption and Income



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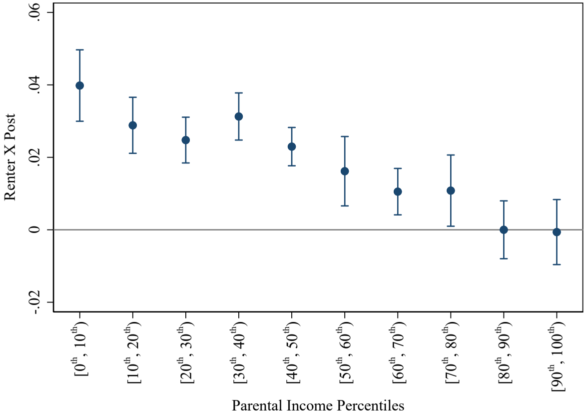
Parental Income Heterogeneity



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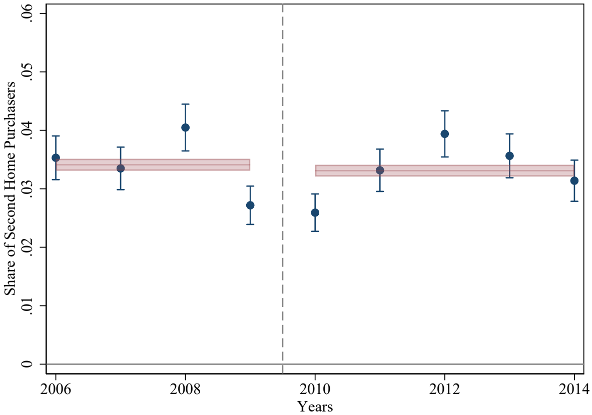
Second Home Purchases of Homeowners



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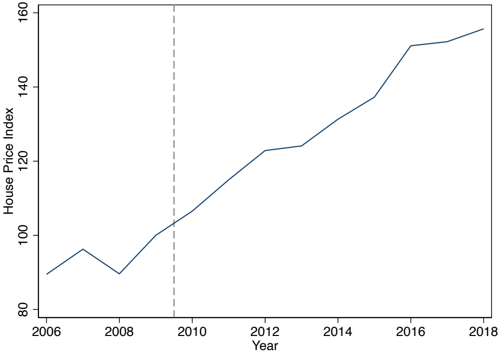
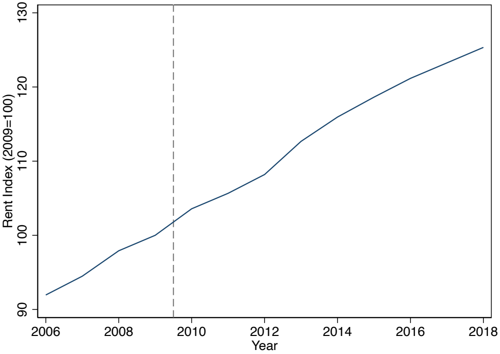
Real Estate Market in Norway



House Price Index



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